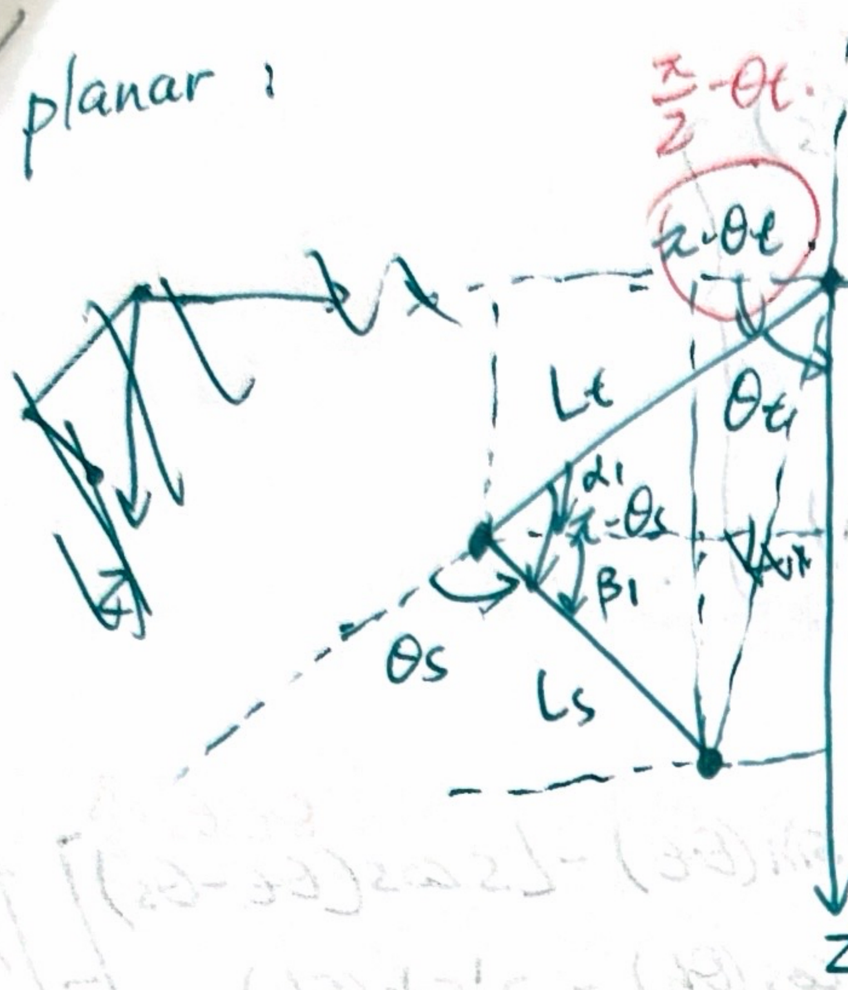
planar :



find x, z ?

$\alpha_1 = \pi - \theta t \cdot \quad \frac{\pi}{2} - \theta t$

$\beta_1 = \pi - \theta s - \alpha_1 \quad \frac{\pi}{2}$

$\beta_1 = \pi - \theta s - \frac{\pi}{2} + \theta t = \boxed{\theta t - \theta s} + \frac{\pi}{2}$

$L t \sin(\theta t)$

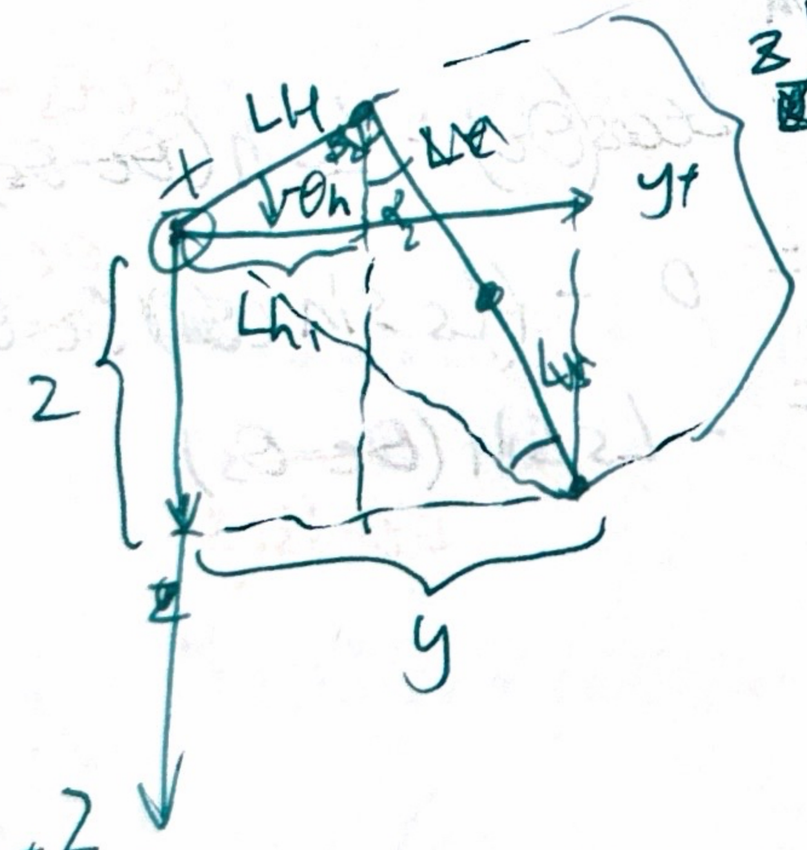
$X t = \theta t \cos L t \cos(\theta t)$

$X s = L s \cdot \cos(\theta t - \theta s) \quad \frac{\pi}{2}$

$\theta t - \theta s + \frac{\pi}{2}$

So $\begin{cases} X = (X t - X s) = L t \cos(\theta t) - L s \cos(\theta t - \theta s) \\ Z' = L t \cos(\theta t) + L s \cdot \sin(\theta t - \theta s) \end{cases}$

Hip :



$L h_1 = L h \cdot \cos(-\theta h) = L h \cos(\theta h)$.

$\beta_2 : \beta_2 + |(-\theta h)| = 180°$

$\begin{cases} |\beta_2| + |\theta h| = \frac{\pi}{2} \\ |\alpha_2| + |\beta_2| = \frac{\pi}{2} \end{cases}$

$\theta h - \alpha_2 = 0. \rightarrow |\theta h = \alpha_2|$

$y = L h_1 + z' \cdot \sin(\theta h) = L h \cos(\theta h) - z' \sin(\theta h)$.

$Z = z' \cos(-\theta h) - LH \sin(-\theta h)$

$Z = z' \cos(\theta h) + LH \sin(\theta h)$.

Fk:

$$\begin{cases} x = L_t \sin(\theta_t) - L_s \cos(\theta_t - \theta_s) \\ y = L_h \cos(\theta_h) - z' \sin(\theta_h) \\ z = z' \cos(\theta_h) + L_h \sin(\theta_h) \end{cases}$$

$\theta_t - \theta_s + \frac{\pi}{2}$

So Fk position transformation A:

$$\vec{\theta} = \begin{bmatrix} \theta_h \\ \theta_t \\ \theta_s \end{bmatrix} \quad \vec{P} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$\vec{P}(\theta_h, \theta_s, \theta_t) = \begin{bmatrix} L_t \sin(\theta_t) - L_s \cos(\theta_t - \theta_s) \\ L_h \cos(\theta_h) - z' \sin(\theta_h) \\ L_h \sin(\theta_h) + z' \cos(\theta_h) \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$\theta_t - \theta_s + \frac{\pi}{2}$

$$z' = L_t \cos(\theta_t) + L_t L_s \sin(\theta_t - \theta_s)$$

differentiate to get Jacobian J:

$$J = \begin{bmatrix} \frac{\partial x}{\partial \theta_h} & \frac{\partial x}{\partial \theta_t} & \frac{\partial x}{\partial \theta_s} \\ \frac{\partial y}{\partial \theta_h} & \frac{\partial y}{\partial \theta_t} & \frac{\partial y}{\partial \theta_s} \\ \frac{\partial z}{\partial \theta_h} & \frac{\partial z}{\partial \theta_t} & \frac{\partial z}{\partial \theta_s} \end{bmatrix} \Rightarrow$$

$$\frac{\partial x}{\partial \theta_h} = 0$$

$$\frac{\partial x}{\partial \theta_t} = L_t \cos(\theta_t) + L_s \sin(\theta_t - \theta_s) \quad \theta_t - \theta_s + \frac{\pi}{2}$$

$$\frac{\partial x}{\partial \theta_s} = 0 - (L_s \sin(\theta_t - \theta_s)) \quad \theta_t - \theta_s + \frac{\pi}{2}$$

$$\frac{\partial y}{\partial \theta_h} = -L_h \sin(\theta_h)$$

$$\frac{\partial y}{\partial \theta_s} = -L_s \sin(\theta_t - \theta_s) \quad \theta_t - \theta_s + \frac{\pi}{2}$$

$$y = L_h \cos(\theta_h) - \left(L_e \cos(\theta_e) \cdot \sin(\theta_h) + L_s \sin(\theta_e - \theta_s) \cdot \sin(\theta_h) \right)$$

$$\frac{\partial y}{\partial \theta_h} = -L_h \sin(\theta_h) - \left(L_e \cos(\theta_e) \cos(\theta_h) + L_s \sin(\theta_e - \theta_s) \cos(\theta_h) \right)$$

$$\frac{\partial y}{\partial \theta_h} = -L_h \sin(\theta_h) - \cos(\theta_h) \cdot \left(L_e \cos(\theta_e) + L_s \sin(\theta_e - \theta_s) \right)$$

$$\frac{\partial y}{\partial \theta_e} = + L_e \sin(\theta_e) \cdot \sin(\theta_h) - L_s \cos(\theta_e - \theta_s) \cdot \sin(\theta_h)$$

$$\frac{\partial y}{\partial \theta_s} = -\left(L_s \cdot \cos(\theta_e - \theta_s) \cdot \sin(\theta_h)(-1) \right) = L_s \cos(\theta_e - \theta_s) \cdot \sin(\theta_h)$$

$$z = L_h \sin(\theta_h) + L_e \cos(\theta_e) \cdot \cos(\theta_h) + L_s \sin(\theta_e - \theta_s) \cdot \cos(\theta_h)$$

$$\frac{\partial z}{\partial \theta_h} = L_h \cos(\theta_h) - L_e \cos(\theta_e) \cdot \sin(\theta_h) - L_s \sin(\theta_e - \theta_s) \sin(\theta_h)$$

$$\frac{\partial z}{\partial \theta_e} = -L_e \sin(\theta_e) \cdot \cos(\theta_h) + L_s \cos(\theta_e - \theta_s) \cdot \cos(\theta_h)$$

$$\frac{\partial z}{\partial \theta_s} = -L_s \cos(\theta_e - \theta_s) \cos(\theta_h)$$

replace all $\theta_e - \theta_s$ with $\theta_e - \theta_s + \frac{\pi}{2}$. !!!

$J =$

$L_e \cos(\theta_e) + L_s \sin(\theta_e - \theta_s)$ $\quad -L_s \sin(\theta_e - \theta_s)$

rque control:

Footend force vector: $F = \begin{bmatrix} F_x \\ F_y \\ F_z \end{bmatrix}$ joint torque: $\vec{\tau} = \begin{bmatrix} \tau_1 \\ \tau_2 \\ \tau_3 \end{bmatrix}$.

for footend to do work: $\delta W = F \cdot \delta p = \begin{bmatrix} F_x \\ F_y \\ F_z \end{bmatrix} \cdot \begin{bmatrix} \delta x \\ \delta y \\ \delta z \end{bmatrix}$

$= \left(\begin{bmatrix} F_x & F_y & F_z \end{bmatrix} \right) \begin{bmatrix} \delta x \\ \delta y \\ \delta z \end{bmatrix} = (F_x \delta x + F_y \delta y + F_z \delta z)$.

$\delta W = F^T \cdot \delta p$. ←

Prove: $\tau = J^T \cdot F$

since: ① $\delta W = F^T \cdot \delta p$.

$P = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$. $\theta = \begin{bmatrix} \theta_1 \\ \theta_2 \\ \theta_3 \end{bmatrix}$.

also $\delta p = J \cdot \delta q$

$\therefore \delta W = F^T \cdot J \cdot \delta q$. ⇒ $F^T \cdot J \cdot \delta q = \tau \cdot \delta q$ ⇒ $\tau = F^T \cdot J$

also $\delta W = \tau \cdot \delta q$

$\delta W = F^T \cdot J \cdot \delta q = (J^T \cdot F)^T \cdot \delta q$

$\therefore \delta W = \tau^T \cdot \delta q$ $\therefore \tau^T \cdot \delta q = (J^T \cdot F)^T \cdot \delta q$

$\tau^T = (J^T \cdot F)^T \Rightarrow \tau = J^T \cdot F$

in quasi-static: $F = J^{-T} \cdot \tau$

↓ inverse of transpose.

④ Impedance Control in Cartesian:

$F_{cmd} = K_p (P_{des} - P_o) + K_d (\dot{P}_{des} - \dot{P}_o)$

↓

$\tau_{cmd} = J^T \cdot F_{cmd}$.

$$J = \begin{bmatrix} 0 & L_t\cos(\theta_t) + L_s\sin(\theta_t - \theta_s) & -L_s\sin(\theta_t \\ -L_h\sin(\theta_h) - \cos(\theta_h)\left(L_t\cos(\theta_t + L_s\sin(\theta_t - \theta_s\right) & L_t\sin(\theta_t)\sin(\theta_h) - L_s\sin(\theta_h)\cos(\theta_t - \theta_s) & L_s\sin(\theta_h)\cos \\ L_h\cos(\theta_h) - \sin(\theta_h)\cdot\left(L_t\cos(\theta_t) + L_s\sin(\theta_t - \theta_s)\right) & L_t\sin(\theta_t)\cos(\theta_h) - L_s\sin(\theta_h)\cos(\theta_t - \theta_s) & -L_s\cos(\theta_h)\cos(\theta_t \\ & -L_s\sin(\theta_t)\cos(\theta_h) + L_s\cos(\theta_t - \theta_s)\cos(\theta_h) & \end{bmatrix}$$

$$J = \begin{bmatrix} 0 & L_t\cos(\theta_t) + L_s\sin(\theta_t - \theta_s) & -L_s\sin(\theta_t - \theta_s) \\ -L_h\sin(\theta_h) - \cos(\theta_h)\cdot\left(L_t\cos(\theta_t) + L_s\sin(\theta_t - \theta_s)\right) & L_t\sin(\theta_t)\sin(\theta_h) - L_s\sin(\theta_h)\cos(\theta_t - \theta_s) & L_s\sin(\theta_h)\cos(\theta_t - \theta_s) \\ L_h\cos(\theta_h) - \sin(\theta_h)\left(L_t\cos(\theta_t) + L_s\sin(\theta_t - \theta_s)\right) & -L_t\sin(\theta_t)\cos(\theta_h) + L_s\cos(\theta_h)\cos(\theta_h - \theta_t) & -L_s\cos(\theta_h)\cos(\theta_t - \theta_s) \end{bmatrix}$$

Joint Velocity vectors: $\dot{q} = \begin{bmatrix} \dot{\theta}_h \\ \dot{\theta}_t \\ \dot{\theta}_s \end{bmatrix}$, foot position vector: $\vec{P} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$.

map from joint velocity → foot end velocity:

$$\vec{\dot{P}} = J \cdot \vec{\dot{q}}$$

① Position control: current position: $\vec{P}_0$, target position: $\vec{P}_t$.

$$\Delta\vec{P}_{error} = \vec{P}_t - \vec{P}_0. \Rightarrow \Delta\vec{P} \approx J(q)\cdot\Delta\vec{q} \quad \text{(local. linear approx)}$$

$$\Delta\vec{q} = J^{-1}\cdot\Delta\vec{P}$$

② velocity control:

$$\vec{\dot{q}} = J^{-1}\cdot\vec{\dot{P}} \quad (\dot{P} = J\cdot\dot{q})$$

near singularity: damped least square IK: $\dot{q} = J^T(JJ^T + \lambda^2 I)^{-1}\cdot\dot{P}$

ed inverse Jacobian:

$$J^{\#} = J^T \cdot (J \cdot J^T + \lambda^2 \cdot I)^{-1}.$$

$$\dot{q}_{cmd} = J^{\#} \cdot \dot{P}_{cmd}. \quad \text{(Joint command)}.$$

$$\dot{q}_{cmd} = J^{\#} \cdot [\dot{P}_{des} + k_p (P_{des} - P_{current})].$$

P_{des} → from foot trajectory.

when homed → neutral standing position.

$P_{current}$ = ~~homed~~ standing position.

$$q_{i+1} = q_i + \dot{q}_i \cdot \delta t.$$

Local Linear approximation using Jacobian + position control.

Jacobian + Position Control Layer

Jacobian + Torque control Layer

goal : output ground Force : F_{des}.

torque control : $\tau_{cmd} = J^T \cdot F_{des}$.

then $I_q = \frac{\tau_{cmd}}{k_t \cdot G\eta}$

G : gear ratio.
η : efficiency
k_t : 0.0616 Nm/A.

Impedance leg control :

$$F_{cmd} = k_p \cdot (P_{des} - P) + k_d (\dot{P}_{des} - \dot{P}_a)$$

P_{des} → from foot trajectory

$\dot{P}_{des}$ → from neutral standing pose.

then $\tau = J^T \cdot F_{des}$.

damped Jacobian explain:

For forward kinematic: let $\vec{p}$ = foot end position, $\vec{q}$ = joint position.

~~if $\vec{P_0} = 0$, then for $\vec{p} = \vec{P_0} + J \cdot \dot{q}$~~

for velocity control: $\dot{p} = J \cdot \dot{q}$, goal:

design a new $J^{\#}$ such that near singularity, $\dot{q}$ and $\dot{p}$ are reasonable.

define lost function: $E(\dot{q}) = \| J\dot{q} - \dot{p} \|^2 + \lambda^2 \| \dot{q} \|^2$.

$$E(\dot{q}, \dot{p}) = (J\dot{q} - \dot{p})^T \cdot (J\dot{q} - \dot{p}) + \lambda^2 \cdot \dot{q}^T \cdot \dot{q}$$

$$= (J\dot{q}^T - \dot{p}^T)(J\dot{q} - \dot{p}) + \lambda^2 \dot{q}^T \cdot \dot{q}$$

$$= (J^T J \dot{q}^T \dot{q} - \dot{p}^T J \dot{q} - J \dot{q}^T \dot{p} + \dot{p}^T \cdot \dot{p}) + \lambda^2 \dot{q}^T \cdot \dot{q}$$

dot produce produce all scalars.

So $\dot{p}^T \cdot J \cdot \dot{q} = J^T \cdot \dot{q}^T \cdot \dot{p}$

$$E(\dot{q}, \dot{p}) = J^T J \dot{q}^T \dot{q} - 2 J^T \dot{q}^T \dot{p} + \dot{p}^T \dot{p} + \lambda^2 \dot{q}^T \dot{q}$$

$$= (\lambda^2 + J^T J) \dot{q}^T \dot{q} - 2 J^T \dot{q}^T \dot{p} + \dot{p}^T \dot{p} \quad \because \dot{q}^T \dot{q} = |\dot{q}|^2$$

$$\frac{\partial E}{\partial \dot{q}} = 2(\lambda^2 + J^T J)\dot{q} - 2 J^T \dot{p} = 0 \text{ (optimize)}$$

$$(\lambda^2 + J^T J)\dot{q} = J^T \dot{p}$$

$$\dot{q} = \frac{J^T \dot{p}}{\lambda^2 + J^T J} = (\lambda^2 + J^T J)^{-1} \cdot J^T \cdot \dot{p} \Rightarrow \boxed{\dot{q} = (\lambda^2 I + J^T J)^{-1} \cdot J^T \cdot \dot{p}}$$

since $\dot{p} = J \cdot \dot{q} \Rightarrow \dot{q} = J^{\#} \dot{p} \Rightarrow \dot{q} = J^{\#} \cdot \dot{p}$

$$\boxed{J^{\#} = (\lambda^2 I + J^T J)^{-1} \cdot J^T}$$

as a result of minimizing velocity error while punish huge velocity move

Summary of Jacobian-Based Legged Robot Control:

1. FK function: $FK(\theta_h, \theta_t, \theta_s) = \begin{bmatrix} L_t \sin(\theta_t) - L_s \cos(\theta_t - \theta_s) \\ L_h \cos(\theta_h) - z' \sin(\theta_h) \\ L_h \sin(\theta_h) + z' \cos(\theta_h) \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \vec{P}$.

$\vec{q} = \begin{bmatrix} \theta_h \\ \theta_t \\ \theta_s \end{bmatrix}$.

2. Jacobian : Map from $\dot{q} \rightarrow \dot{p}$:

replace all $\theta_t - \theta_s$ with $\theta_t - \theta_s + \frac{\pi}{2}$!!!

$$J = \begin{bmatrix} 0 & L_t\cos(\theta_t) + L_s\sin(\theta_t - \theta_s + \frac{\pi}{2}) & -L_s\sin(\theta_t - \theta_s + \frac{\pi}{2}) \\ L_h\sin(\theta_h) - \cos(\theta_h)\cdot\left(L_t\cos(\theta_t) + L_s\sin(\theta_t - \theta_s + \frac{\pi}{2})\right) & L_t\sin(\theta_t)\sin(\theta_h) - L_s\sin(\theta_h)\cos(\theta_t - \theta_s + \frac{\pi}{2}) & L_s\sin(\theta_h)\cos(\theta_t - \theta_s + \frac{\pi}{2}) \\ L_h\cos(\theta_h) - \sin(\theta_h)\left(L_t\cos(\theta_t) + L_s\sin(\theta_t - \theta_s + \frac{\pi}{2})\right) & -L_t\sin(\theta_t)\cos(\theta_h) + L_s\cos(\theta_h)\cos(\theta_t - \theta_s + \frac{\pi}{2}) & L_s\cos(\theta_h)\cos(\theta_t - \theta_s + \frac{\pi}{2}) \end{bmatrix}$$

3. Position control loop (joint position control):

defined damped Jacobian (~~ratio~~ minimizing lose function $E(\dot{q}, \dot{p})$ for velocity)

$$J^{\#} = J^T \cdot (J \cdot J^T + \lambda^2 I)^{-1}$$

① Home all legs: $\vec{P}_0 = \vec{P}_{home}$, $\vec{\ell}_0 = \vec{\ell}_{home}$.

② Define foot trajectory: $\vec{P}_{desired} = traj(x, y, z)$ (incremental)

③ Compute $\dot{P}_{des} = \frac{\vec{P}_{des}}{\Delta time} \rightarrow \Delta time = \frac{1}{f_{loop}}$. $f_{loop} \approx 80 \sim 120Hz$ (or higher).

④. compute $\dot{q}_{des} = J^{\#} \cdot \dot{P}_{des} + k_p(\vec{P}_{des} - \vec{P}_i)$

P_i = updated encoder reading each loop, convert into leg cartesian space. (not raw reding). $\vec{P}_0 = \vec{P}_{home}$.

⑤ $q_{i+1} = q_i + \dot{q}_i \cdot \Delta time$

↓

output incremental joint position ~~displacement~~ command to all joints.